

Project NotebookSF

Technical Document Specification - Version 1.0

Overview:

NotebookSF is a research and note-taking assistant similar to Google's NotebookLM. A key differentiator for this application is the ability to organize areas of knowledge in a hierarchical fashion using the concept of Groups, Projects and Notebooks. Once a notebook is defined, users have the added ability to categorize information into sections.

Each notebook section will have the following four 'Interaction Modes':

1. **Resources:** An Interaction Mode that serves as a collection of referenced URL links with their respective page title. Resources can be clicked to navigate users to the referenced URL. Resources can be created, renamed or deleted.
2. **Notes:** An Interaction Mode that serves as a collection of rich formatted notes taken by the user. Notes can be created, renamed or deleted.
3. **Cards:** An Interaction Mode that serves as a collection of flippable flashcards used for learning. They are flippable with a question on the front and an answer on the back. These can be created using generative AI or manually entered. Cards can be created, renamed or deleted.
4. **Exams:** An Interaction Mode that serves as a collection of topic related exam questions generated from the list of Resources and Notes used for learning. Exams can be created manually or via generative AI, renamed or deleted.

To avoid 'category clutter', two features will be available:

- **Starred Items:** Each 'Interaction Mode' will consist of individual items that can be starred. Starring an item will place it into the notebook's designated 'Starred' section for easy reference.
- **AI Chat Component:** This will leverage Retrieval Augmented Generation to search, find and provide summarized answers across all sections, notebooks, projects and groups.

Technology Stack:

- Platform: Salesforce, API version 66 (latest)
- Backend Logic: Apex
- Data Storage: Custom Objects
- UI Framework: Lightning Console Application using Lightning Component Tabs + Lightning Web Components
- CSS Theme: Salesforce Cosmos
- Audience: System Administrators profile

User Stories:

- **Add Group:** As a user, I want to have the ability to add a new group from the 'Application Home Page' for organizing a collection of related projects. This will display the 'Create / Edit Group' modal.
- **Edit Group:** As a user, I want to have the ability to rename an existing group from the 'Application Home Page' of organized collections of projects. This will display the 'Create / Edit Group' modal. The group's edit button will be visible only when hovering over the group's title.
- **Delete Group:** As a user, I want to have the ability to delete an existing group from the 'Application Home Page' of organized projects of related notebooks. This will display the 'Delete Group' confirmation modal. The group's delete button will be visible only when hovering over the group title, but only if it contains no projects.
- **Select Group:** As a user, I want to have the ability to select an existing group from the 'Application Home Page' to display a collection of projects of related notebooks.

- **Add Project:** As a user, I want to have the ability to add a new project from the 'Application Home Page' for organizing a collection of notebooks.
- **Edit Project:** As a user, I want to have the ability to rename or reparent an existing project from the 'Application Home Page'. This will display the 'Create / Edit Project' modal. The group's edit button will be visible only when hovering over the group's title.
- **Delete Project:** As a user, I want to have the ability to delete an existing project from the 'Application Home Page'. This will display the 'Delete Project' confirmation modal. The project's delete button will be visible only when hovering over the project title, but only if it contains no notebooks.

- **Add Notebook:** As a user, I want to have the ability to add a new notebook from the 'Application Home Page' for accessing the notebook's interaction modes.
- **Edit Notebook:** As a user, I want to have the ability to rename or reparent an existing notebook from the 'Application Home Page'. This will display the 'Create / Edit Notebook' modal. The notebook's edit button will be visible only when hovering over the notebook's title.
- **Delete Notebook:** As a user, I want to have the ability to delete an existing notebook from the 'Application Home Page'. This will display the 'Delete Notebook' confirmation modal. The notebook's delete button will be visible only when hovering over the project title. The confirmation modal should emphasize that all contents within the notebook will be deleted.
- **Select Notebook:** As a user, I want to have the ability to select an existing notebook from the 'Application Home Page' which will render the 'Resources Interaction Mode - List View' page.

- **Add Section:** As a user, I want to have the ability to add a new section for categorizing items from the 'Resources Interaction Mode - List View' page. This will render the 'Create / Edit Section' modal.
- **Edit Section:** As a user, I want to have the ability to rename an existing section title from the 'Resources Interaction Mode - List View' page. This will display the 'Create / Edit Section' modal. The section's edit button will be visible only when hovering over the section's title.
- **Delete Section:** As a user, I want to have the ability to delete an existing section from the 'Resources Interaction Mode - List View' page. This will display the 'Delete Section' confirmation modal. The section's delete button will be visible only when hovering over the section title. The confirmation modal should emphasize that all contents within the section will be deleted.

- **Select Resource Interaction Mode:** As a user, I want the ability to select the 'Resources' Interaction Mode which will render the 'Resources Interaction Mode - List View' page. This will be available via the 'Resources' interaction mode button near the top of the page.
- **Select Resource:** As a user, I want to have the ability to select an existing resource from the 'Resources Interaction Mode - List View' page which will open a new browser tab and navigate to the resource's associated URL.
- **Add Resource:** As a user, I want to have the ability to add a new resource from the 'Resources Interaction Mode - List View' page. This will render the 'Create / Edit Resource' modal.
- **Edit Resource:** As a user, I want to have the ability to edit an existing resource from the 'Resources Interaction Mode - List View' page so I can edit the resource link and title.
- **Delete Resource:** As a user, I want to have the ability to delete an existing resource from the 'Resources Interaction Mode - List View' page. This will display the 'Delete Resource' confirmation modal. The resource's delete button will be visible only when hovering over the resource title. The confirmation modal should emphasize that all contents associated with the resource will be lost.

- **Select Note Interaction Mode:** As a user, I want the ability to select the 'Notes' Interaction Mode which will render the 'Notes Interaction Mode - List View' page. This will be available via the 'Notes' interaction mode button near the top of the page.

- **Select Note:** As a user, I want to have the ability to select an existing note from the 'Notes Interaction Mode - List View' page. This will render the 'Notes Interaction Mode - Details View' page.
- **Create Note:** As a user, I want to have the ability to create a new note from the 'Notes Interaction Mode - List View' page. This will render the 'Create or Edit Note' modal
- **Delete Note:** As a user, I want to have the ability to delete an existing note from the 'Notes Interaction Mode - List View' page.
- **Close Note:** As a user, I want to have the ability to save and close an existing note from the 'Notes Interaction Mode - Details View' page. This will navigate the user to the 'Notes Interaction Mode - List View' page. This will be available by clicking the 'Save & Close' button near the bottom of the page.
- **Cancel Note Edit:** As a user, I want to have the ability to cancel any edits to an existing note from the 'Notes Interaction Mode - Details View' page. This will navigate the user to the 'Notes Interaction Mode - List View' page. This will be available by clicking the 'Cancel' button near the bottom of the page.
- **Select Card Interaction Mode:** As a user, I want to have the ability to select an existing note from the 'Cards Interaction Mode - List View' page. This will be available via the 'Cards' interaction mode button near the top of the page.
- **Open Deck:** As a user, I want to have the ability to select an existing card deck from the 'Cards Interaction Mode - List View' page which will render the 'Cards Interaction Mode - Details View' page which displays each card one at a time in a carousel type fashion.
- **Add Card:** As a user, I want to have the ability to add a new flipcard from the 'Cards Interaction Mode - Details View' page for displaying a topic's question & answer learning content.
- **Edit Card:** As a user, I want to have the ability to edit an existing flipcard from the 'Cards Interaction Mode - Details View' page for updating the front and back content of the card. This will render the first of the 'Cards Interaction Mode - Upsert Series' pages.
- **Delete Card:** As a user, I want to have the ability to delete an existing card from the 'Cards Interaction Mode - Details View' page.
- **Sort Cards:** As a user, I want to have the ability to specify the order of cards from the 'Cards Interaction Mode - Details View' page to display them in a specified sequence.
- **Navigate to Next Card:** As a user, I want to have the ability to navigate to the next card from the 'Cards Interaction Mode - Details View' page. This can be done by clicking the right arrow button near the bottom of the page.
- **Navigate Previous to Card:** As a user, I want to have the ability to navigate to the previous card from the 'Cards Interaction Mode - Details View' page. This can be done by clicking the left arrow button near the bottom of the page.
- **Flip Card:** As a user, I want to have the ability to flip the card from the 'Cards Interaction Mode - Details View' page to view the card's other side content. This can be done by either clicking the card or pressing the spacebar.
- **Close Deck:** As a user, I want to have the ability to close the card deck from the 'Cards Interaction Mode - Details View' page. This will navigate the user to the 'Cards Interaction Mode - List View' page.
- **Select Exam Interaction Mode:** As a user, I want to have the ability to select an existing exam from the 'Exam Interaction Mode - List View' page. This will be available via the 'Exam' interaction mode button near the top of the page.
- **Open Exam:** As a user, I want to have the ability to select an existing exam from the 'Exam Interaction Mode - List View' page which will render the 'Exam Interaction Mode - Details View' page which displays each exam question one at a time.
- **Create Exam Question:** As a user, I want to have the ability to add an existing exam question from the 'Exam Interaction Mode - Details View' page for creating the exam question, answers and explanation. This will render the first of the 'Exam Interaction Mode - Upsert Series' pages.
- **Edit Exam:** As a user, I want to have the ability to edit an existing exam question from the 'Exam Interaction Mode - Details View' page for updating the exam question, answers and explanation. This will render the first of the 'Exam Interaction Mode - Upsert Series' pages.
- **Delete Exam:** As a user, I want to have the ability to delete an existing exam question from the 'Exam Interaction Mode - Details View' page.
- **Close Exam:** As a user, I want to have the ability to close the exam from the 'Exam Interaction Mode - Details View' page. This will navigate the user to the 'Exam Interaction Mode - List View' page.

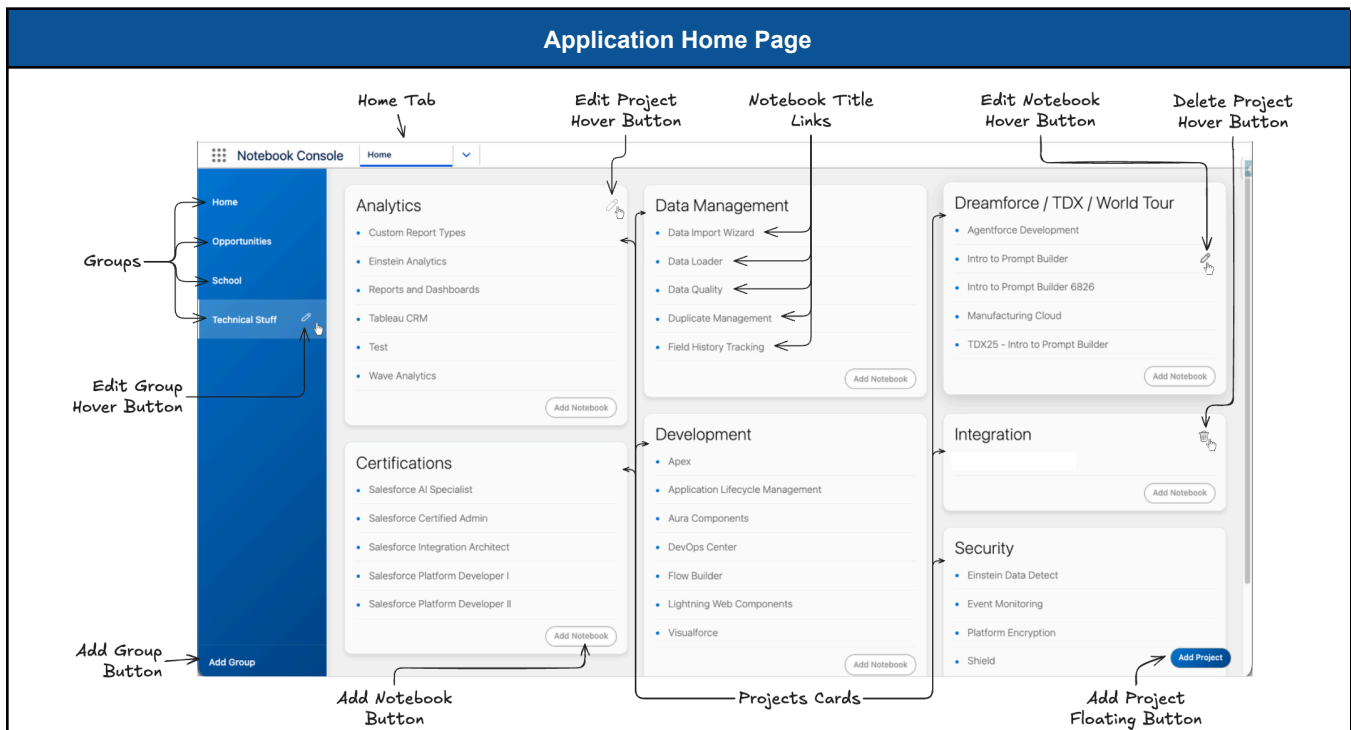
User Interface:

The user interface consists of several screen pages including:

- Application Home Page
- Resources Interaction Mode - List View
- Resources Interaction Mode - Detail View
- Notes Interaction Mode - List View
- Notes Interaction Mode - Detail View
- Cards Interaction Mode - List View
- Cards Interaction Mode - Detail View
- Cards Interaction Mode - Upsert Series (Page 1 of 3)
- Cards Interaction Mode - Upsert Series (Page 2 of 3)
- Cards Interaction Mode - Upsert Series (Page 3 of 3)
- Cards Interaction Mode - Card Sort View
- Exams Interaction Mode - List View
- Exams Interaction Mode - Detail View
- Exam Interaction Mode - Upsert Series (Page 1 of 2)
- Exam Interaction Mode - Upsert Series (Page 2 of 2)
- Exam Interaction Mode - Question Sort View

Each screen, except for the 'Application Home Page' will have an AI Chat component fixed to the right side of the screen. In addition to the screen pages mentioned above, this application will also have several modals and confirmation modals using the 'lwcModal' component included in the project. Details and sketches of these pages and components are outlined in the following pages.

Use the following sketch designs as suggestions. Adjust the UI as necessary to abide by best practices and keep the interface visually appealing.



This is the initial application page. On the far left is the “Groups’ navigation menu. The primary focus for the ‘Application Home Page’ is the collection of ‘Project’ cards associated with the selected ‘Group’. Each “Project’ card contains a list of ‘Notebook’ items. From this screen the user can create a new ‘Group’, create a new ‘Project’ and create a new ‘Notebook’. Below are the modals for this page:

The ‘Create / Edit Group’ modal:

The ‘Delete Group’ confirmation modal:

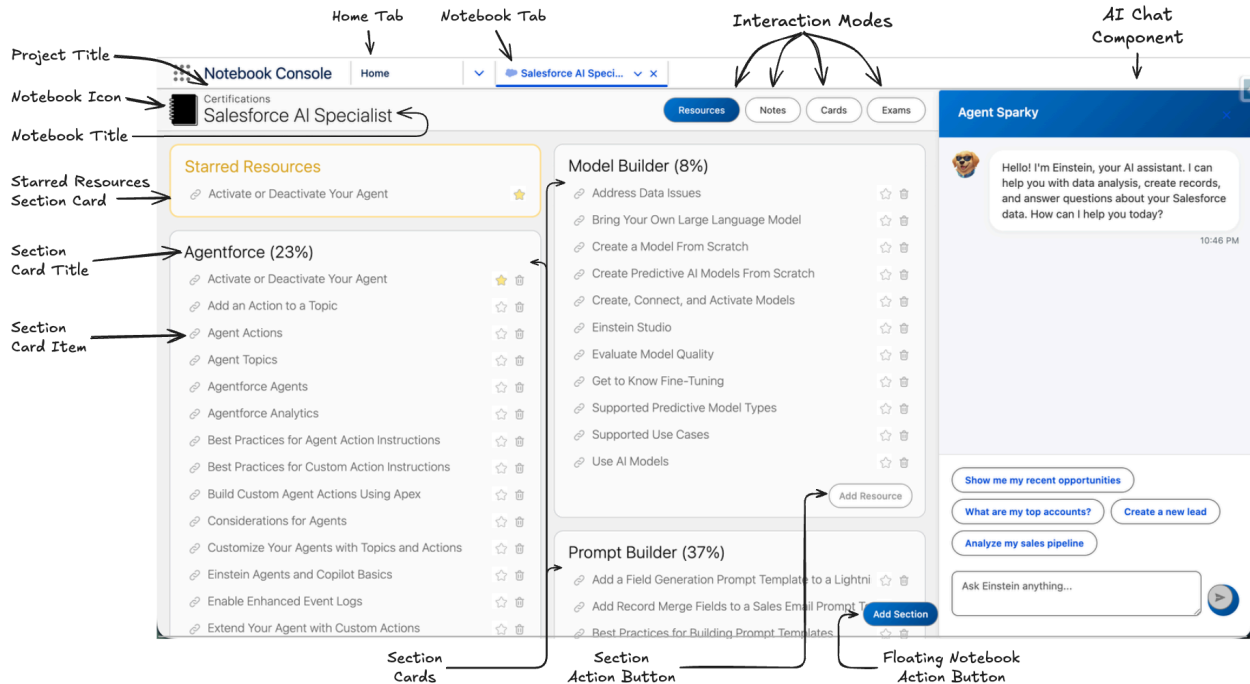
The ‘Create / Edit Project’ modal:

The ‘Delete Project’ confirmation modal:

The ‘Create / Edit Notebook’ modal:

The ‘Delete Notebook’ confirmation modal:

Resources Interaction Mode - List View



Clicking a 'Notebook Title Link' from the 'Application Home Page' or clicking the 'Resources' interaction mode button will display the 'Resources Interaction Mode - List View' page. The main portion consists of one or more 'Section Cards', each displaying the section's title and one or more 'Section Card Items'. Each 'Section Card Item' will have an 'Item Edit Hover Button' (for displaying the 'Create / Edit Resource Modal' to edit the resource's title and link), an 'Item Star Toggle Button' (for toggling that item to and from the 'Starred Resources Section Card') and a 'Item Delete Button' (for deleting that item). Each 'Section Card' (with the exception of the 'Starred Resources Section Card') will have a 'Section Action Button' labeled 'Add Resource Button' in the bottom right corner of that 'Section Card' for displaying the 'Create / Edit Resource Modal'. Clicking on a 'Section Card Item' will open a new tab and navigate the user to the resource's URL link. On the far right is the AI chat panel containing the Fetch Agent. In the bottom right corner of the page will be a 'Floating Notebook Action Button' labeled 'Add Section' for adding new sections to the notebook by calling the 'Create / Edit Section' modal. The main purpose of this interaction mode is to provide an interface for organizing resources about a specific topic related to the notebook. Below are the modals for this page:

The 'Create / Edit Section' modal:

The 'Create / Edit Section' modal is a white box with a dark header. It contains a text input field labeled '* Enter a name for the new section:' and two buttons at the bottom: 'Cancel' and 'Create'.

The 'Delete Section' confirmation modal:

The 'Delete Section' confirmation modal is a white box with a dark header. It contains a text input field labeled 'Delete this Section?' and a message: 'Click OK to continue with deleting this section. Otherwise, click Cancel.' There are two buttons at the bottom: 'Cancel' and 'OK'.

The 'Create / Edit Resource' modal:

The 'Create / Edit Resource' modal is a white box with a dark header. It contains two text input fields: '* Resource Link:' and '* Title:'. Below the 'Resource Link' field are three buttons: 'Link', 'PDF', and 'Video'. At the bottom are two buttons: 'Cancel' and 'Create'.

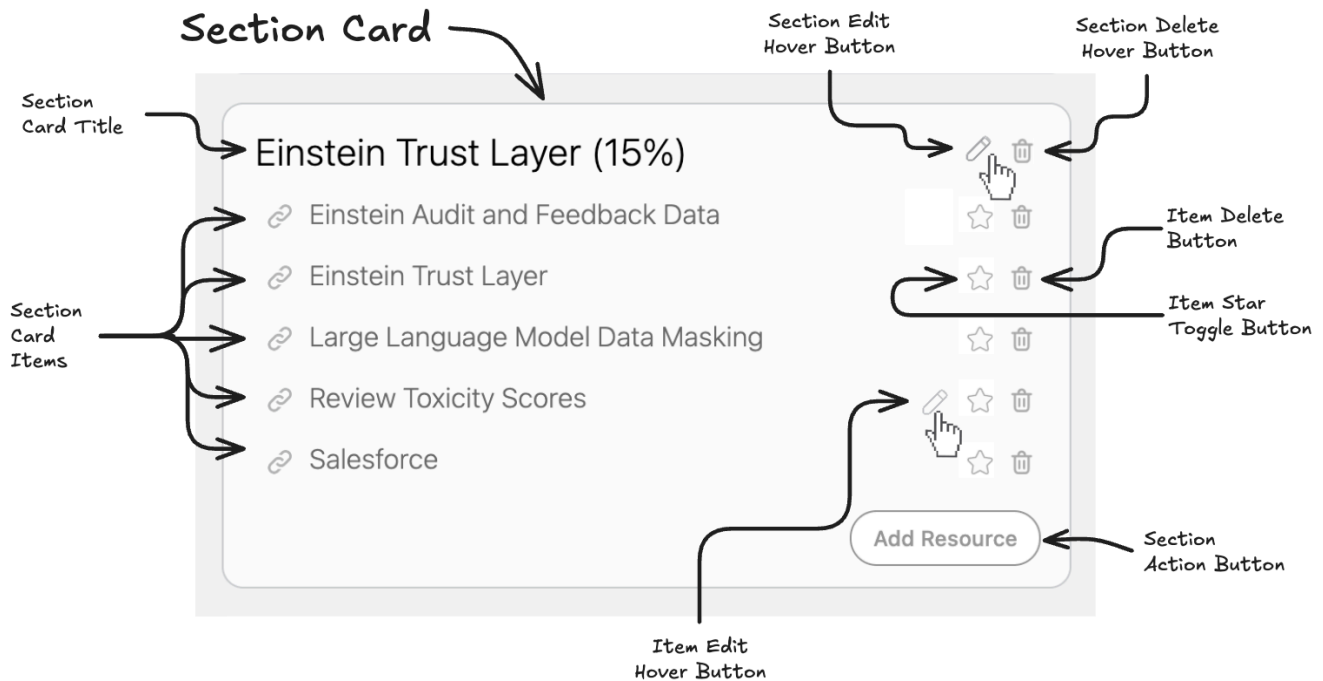
The 'Delete Resource' confirmation modal:

The 'Delete Resource' confirmation modal is a white box with a dark header. It contains a text input field labeled 'Delete this resource?' and a message: 'Deleting this resource will also delete all content associated with it. This includes notes, review cards, exam questions, etc. Click OK to continue, otherwise, click Cancel.' There are two buttons at the bottom: 'Cancel' and 'OK'.

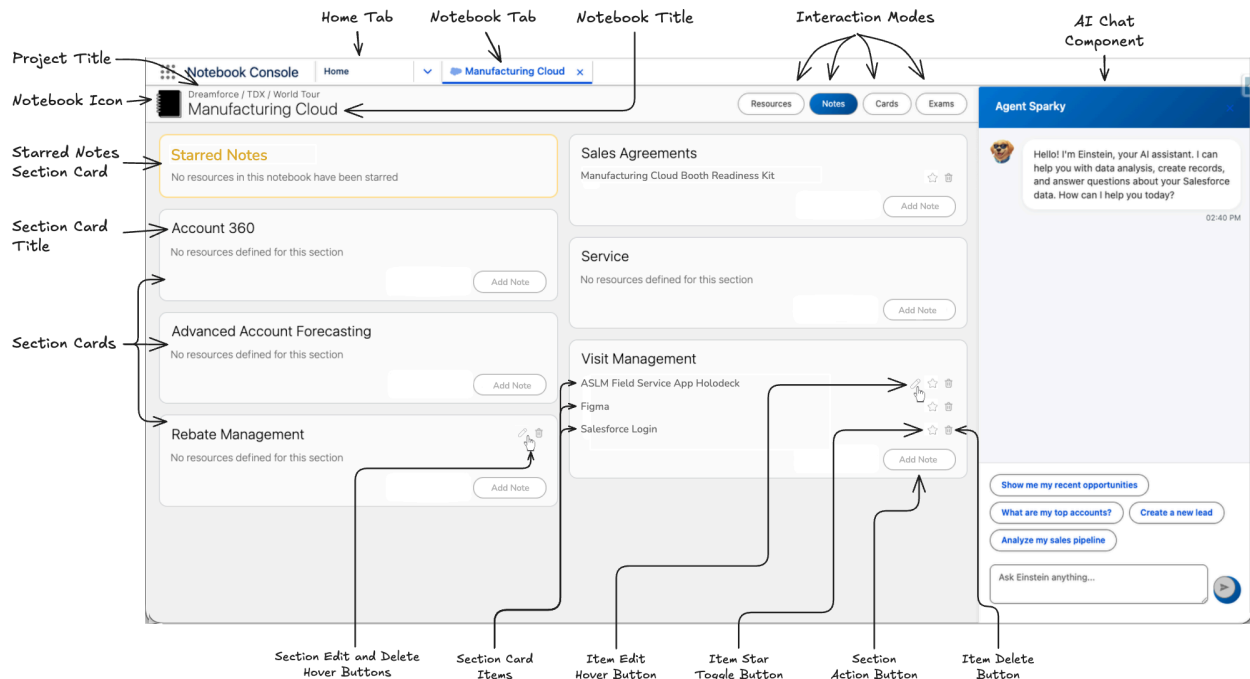
List View Interaction Mode Pages - Special Notes

Each of the 'List View' pages (Resources, Notes, Cards & Exams) consist of a series of 'Section Cards' displayed in a columnar fashion. These 'Section Cards' represent individual areas of knowledge within a notebook, each consisting of a single section card title, one or more section card items and one or more section action buttons with the following characteristics:

1. **Section Card Title:** These will have edit and delete icons appear when hovering.
2. **Section Card Items:** These will have an edit icon that appears when hovering. Two additional icons will also be continuously displayed: a 'starred' toggle icon and a 'delete' icon.
3. **Section Action Buttons:** This will be one or more buttons to appear in the lower right corner of the Section UI Card.



Notes Interaction Mode - List View



Clicking the 'Notes' interaction mode button displays the 'Notes Interaction Mode - List View' page. The main portion consists of one or more 'Section Cards', each displaying the section's title and one or more 'Section Card Items'. Each 'Section Card Items' will have a 'Item Edit Hover Button' (for displaying the 'Create / Edit Note Modal' to edit the note's title), a 'Item Star Toggle Button' (for toggling that item to and from the 'Starred Resources' section) and a 'Item Delete Button' (for deleting that item). Each 'Section' card (with the exception of the 'Starred Notes Section Card') will have a 'Section Action Button' labeled 'Add Note Button' in the bottom right corner of that card for displaying the 'Create / Edit Note Modal'. Upon creation, the new note's title will be defined and the user will be taken to the 'Notes Interaction Mode - Details View' page. Clicking on a 'Section Card Item' will also navigate the user to the 'Notes Interaction Mode - Details View' page. On the far right is the AI chat panel containing the Fetch Agent. The main purpose of this interaction mode is to provide an interface for organizing notes about a specific topic related to the notebook. Below are the modals for this page:

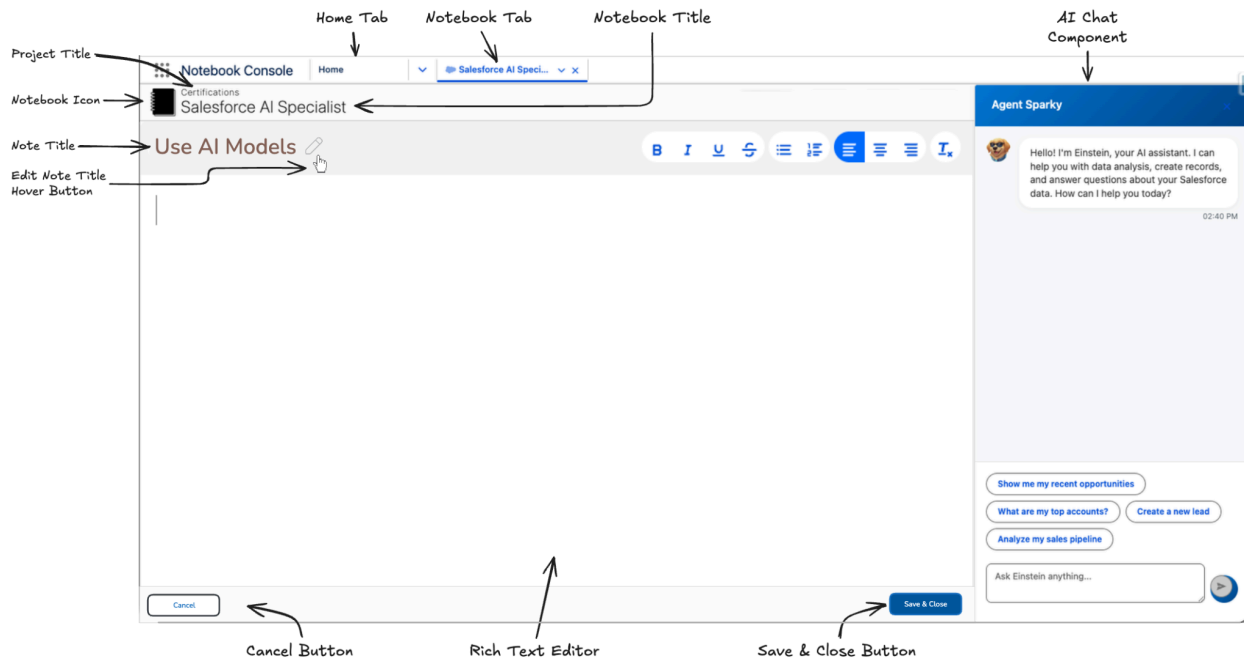
The 'Create / Edit Note' modal:

The 'Create Note' modal is a light gray box with a title bar 'Create Note'. It contains two input fields: '* Enter a title for the new note:' and '* Section:'. The 'Section' dropdown is currently set to 'ASLM Field Service App Handbook'. At the bottom, there are 'Cancel' and 'Create' buttons.

The 'Delete Note Confirmation' modal:

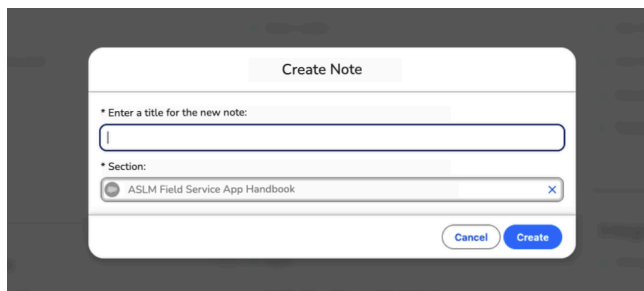
The 'Delete this Note?' modal is a light gray box with a title bar 'Delete this Note?'. It contains a message: 'Click OK to continue with deleting this note. Otherwise, click Cancel.' At the bottom, there are 'Cancel' and 'OK' buttons.

Notes Interaction Mode - Details View

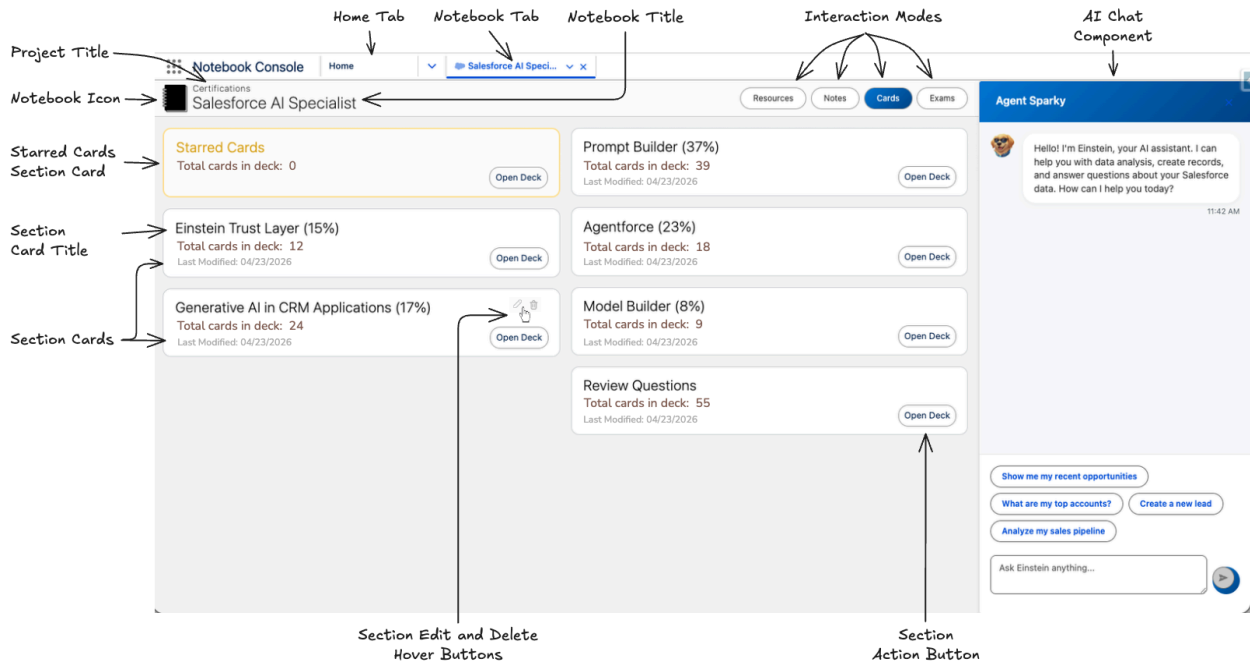


The 'Notes Interaction Mode - Details View' page primarily consists of a rich text editor for taking notes. The top of the page will display the Note's title. Hovering over the note title will enable an edit icon that can be clicked to display the 'Create / Edit Note' modal. The bottom of the page will contain a 'Cancel' button and a 'Save and Close' button for saving any modifications made to the note. Both the 'Cancel' and 'Save and Close' button will navigate the user back to the 'Notes Interaction Mode - List View' page. On the far right is the AI chat panel containing the Fetch Agent.

The 'Create / Edit Note' modal:



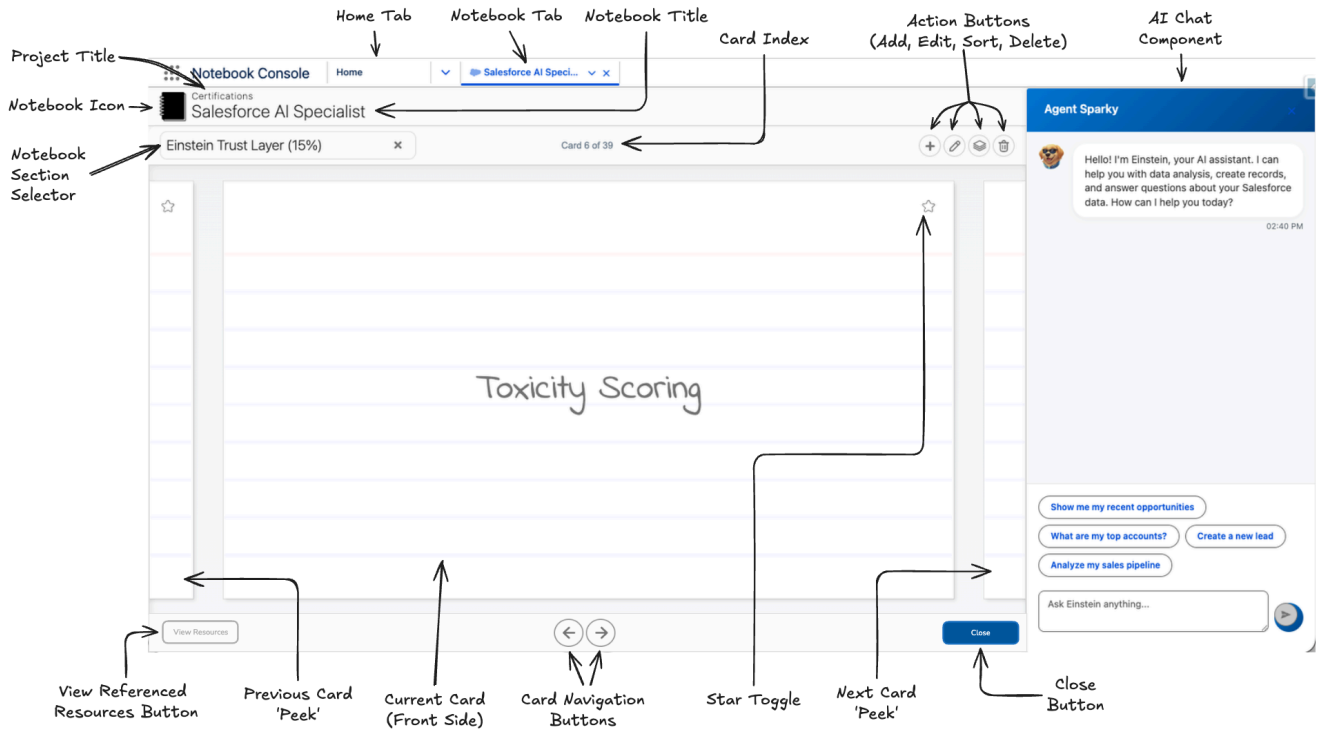
Cards Interaction Mode - List View



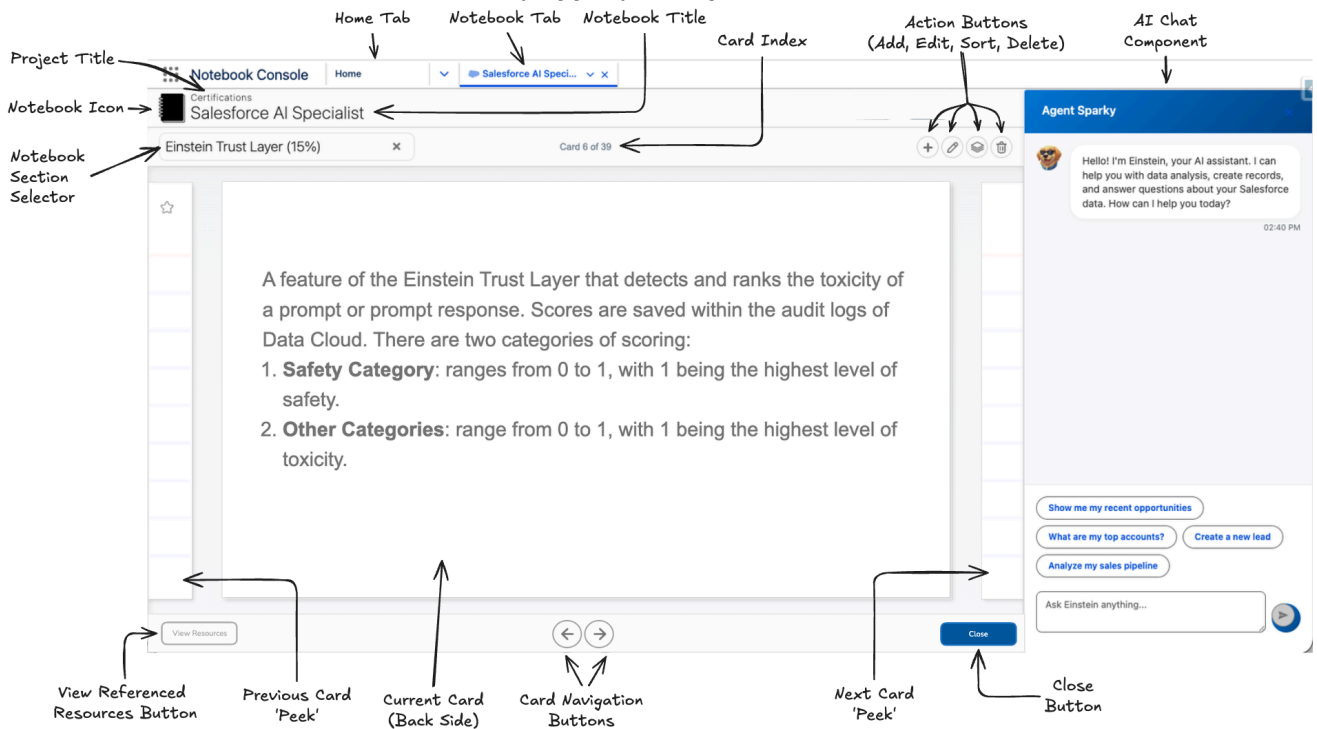
Clicking the 'Cards' interaction mode button displays the 'Cards Interaction Mode - List View' page. The main portion consists of one or more 'Section Cards', each displaying the section's title along with details of the deck such as number of cards and the date last modified. Each 'Section Card' (including the 'Starred Cards Section Card') will have an 'Open Deck Button' in the bottom right corner of that card. Clicking the 'Open Deck' button will navigate the user to the 'Cards Interaction Mode - Details View' page. On the far right is the AI chat panel containing the Fetch Agent. The main purpose of this interaction mode is to provide an interface for organizing cards about a specific topic related to the notebook. There are no modals for this page.

Cards Interaction Mode - Details View

Frontside (Unflipped) Example of Flashcard:



Backside (Flipped) Example of Flashcard:



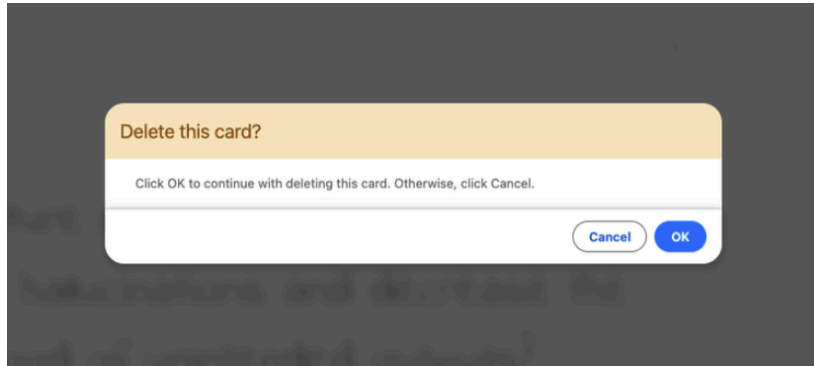
The 'Cards Interaction Mode - Details View' displays flashcards one at a time, which are two-sided study tools used for memorizing information through active recall, featuring a question, prompt, or term on one side and the corresponding answer or definition on the other. Clicking the card, or pressing the space bar, will flip the card. The top of the page will display the notebook's section selector, allowing the user to jump to another section easily. To the right of the notebook section selector are four action buttons: Add (adding a new card), Edit (editing an existing card), Sort (defining the sort order of the cards) and Delete (for deleting a card). It is important the user can access supporting resources for reference. The bottom of the page will contain a 'View

Resources' button which displays a list of all associated sources. A 'Close' button on the lower right will allow the user to close this view and go back to the 'Cards Interaction Mode - List View' page. Finally, the bottom center will have two buttons for navigating through the deck of cards. Prior to navigating, the component will make sure the current card is in the 'unlipped' state. On the far right side is the AI chat panel containing the Fetch Agent.

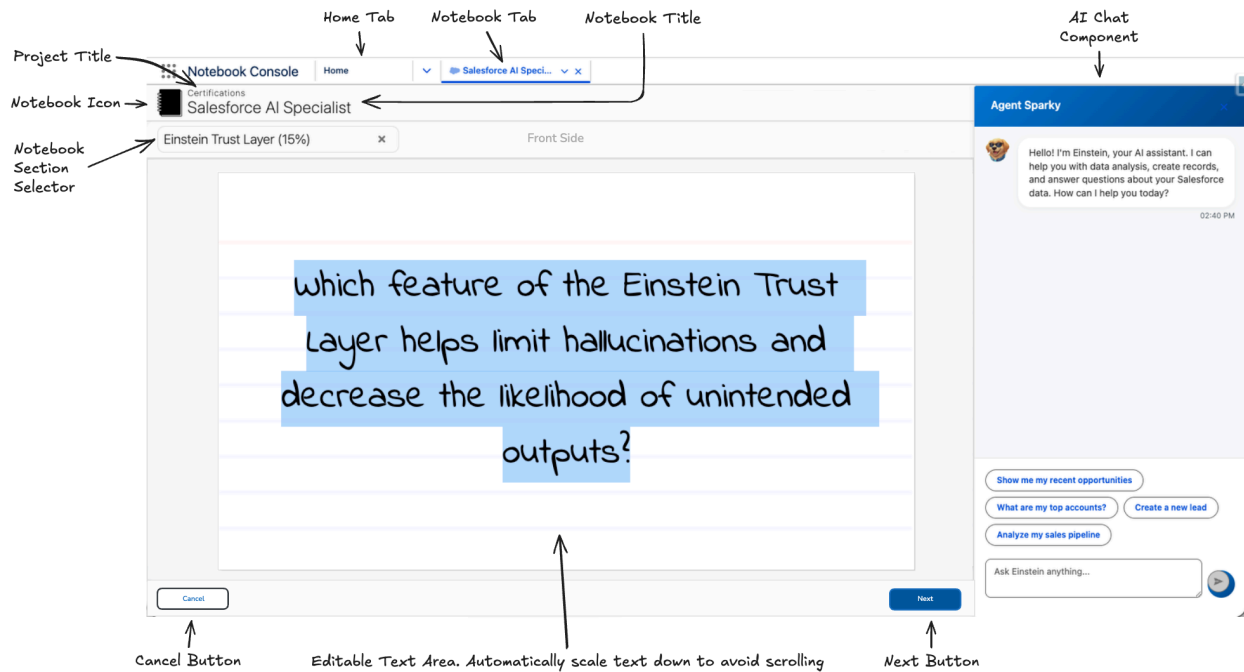
Clicking the 'Add' or 'Edit' action buttons will initiate the 'Cards Interaction Mode - Upsert View' series of pages.

Clicking the 'Sort' action button will initiate the 'Cards Interaction Mode - Card Sort View' page.

Clicking the 'Delete' action button will initiate the 'Delete Card Confirmation' modal shown below:

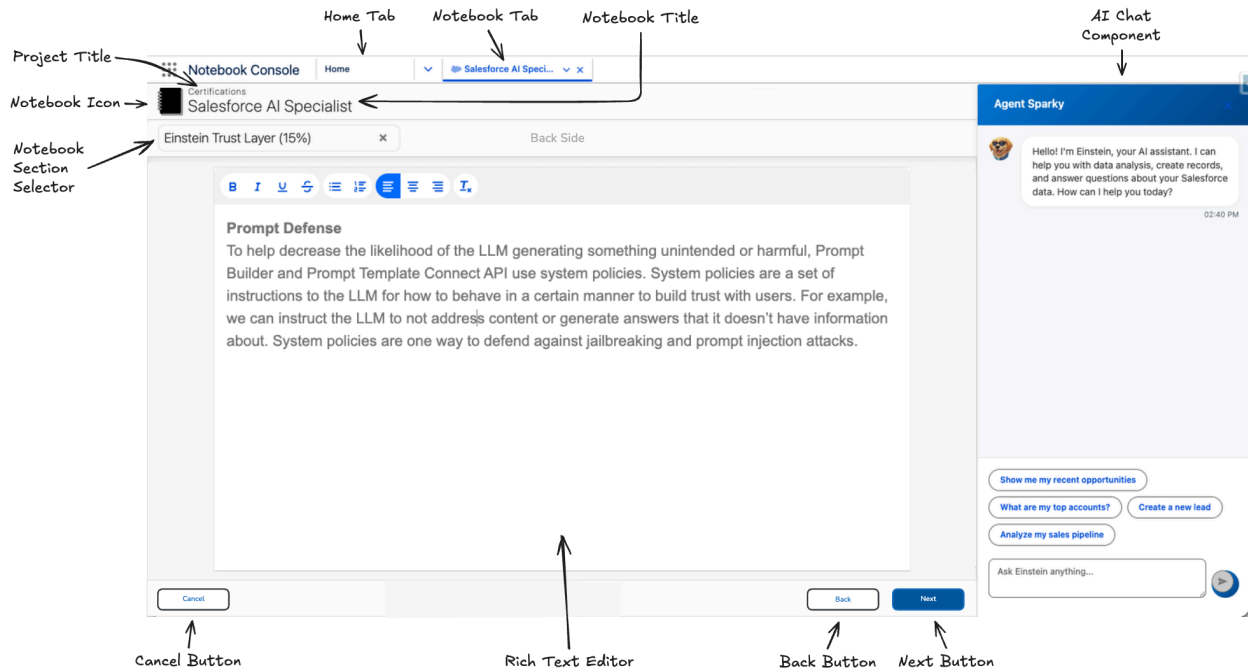


Cards Interaction Mode - Upsert Series (Page 1 of 3)



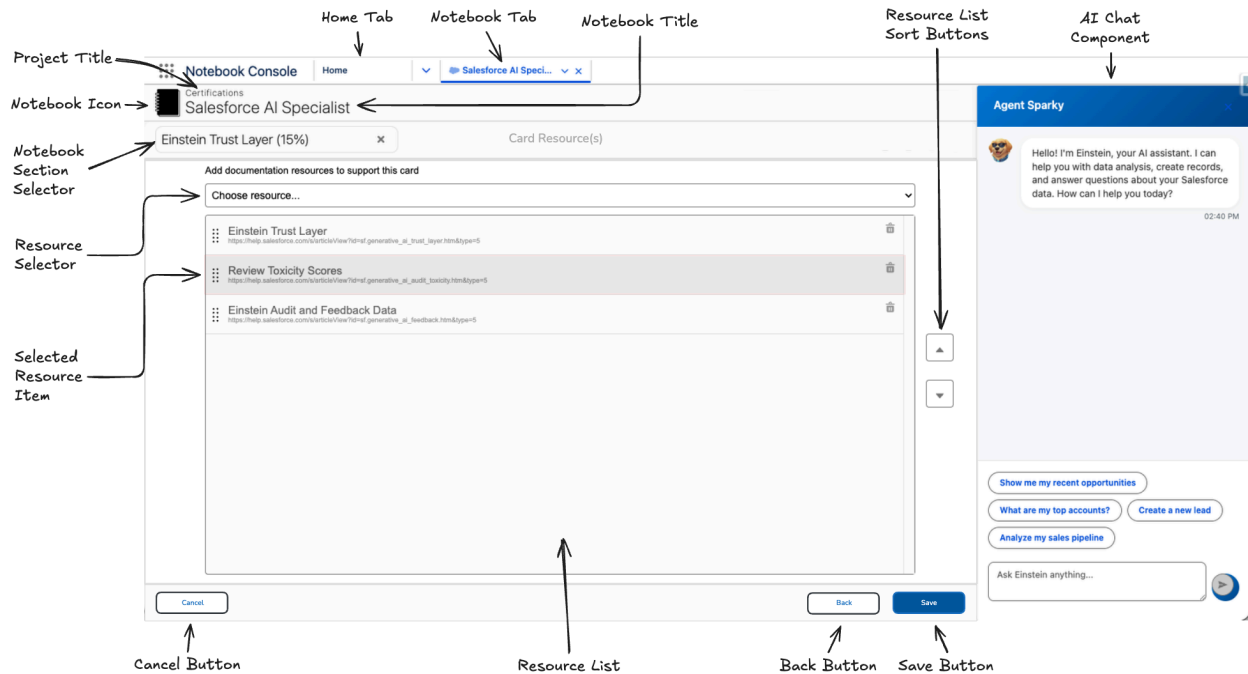
This is the first page of the 'Cards Interaction Mode - Upsert' page series. This screen captures the question, prompt, or term to be displayed on the front side of the flashcard. Text size should automatically scale down to fit content without scrolling. Clicking the 'Cancel' button will cancel adding/updating the card and navigate the user back to the 'Cards Interaction Mode - Details View' page. Clicking the 'Next' button will navigate the user to the next page of the 'Cards Interaction Mode - Upsert Series (Page 2 of 3)' page series.

Cards Interaction Mode - Upsert Series (Page 2 of 3)



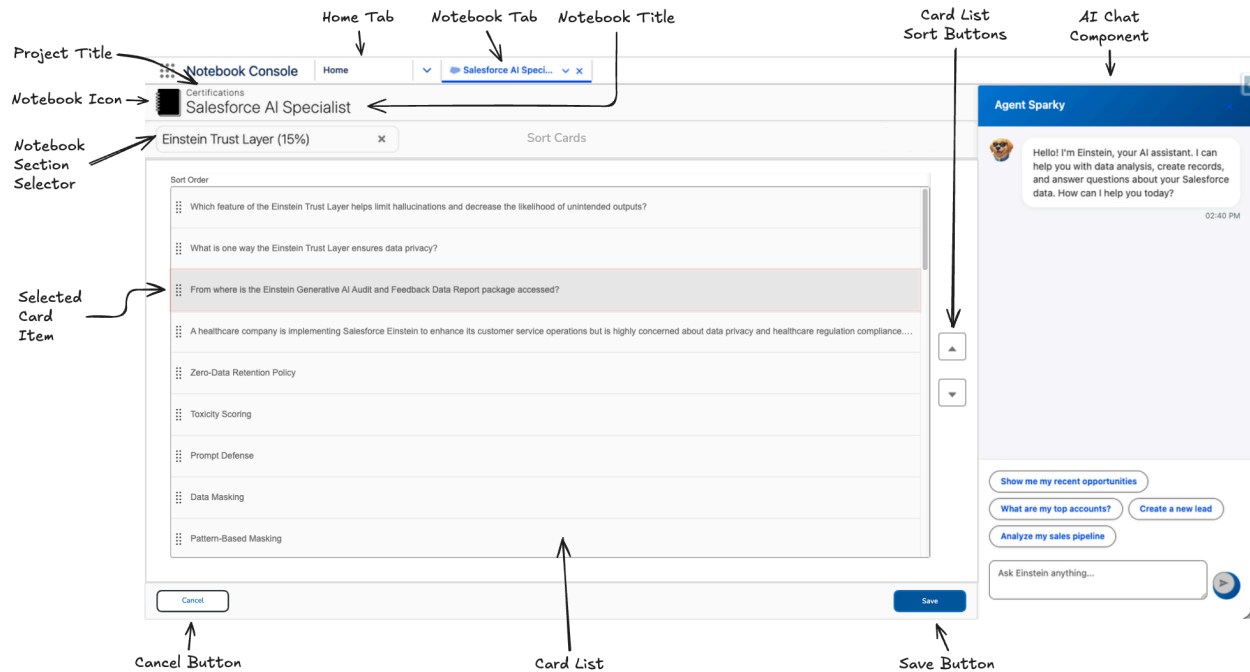
This is the second page of the 'Cards Interaction Mode - Upsert' page series. The purpose of this screen is to capture the question, prompt, or term to be displayed on the front side of the flashcard. The font size should automatically scale down to fit all content within the card without scrolling. Clicking the 'Cancel' button will cancel adding/updating the card and navigate the user back to the 'Cards Interaction Mode - Details View' page. Clicking the back button will navigate the user to the previous page of this upsert series. Clicking the 'Next' button will navigate the user to the next page of the 'Cards Interaction Mode - Upsert Series (Page 2 of 3)' page series.

Cards Interaction Mode - Upsert Series (Page 3 of 3)



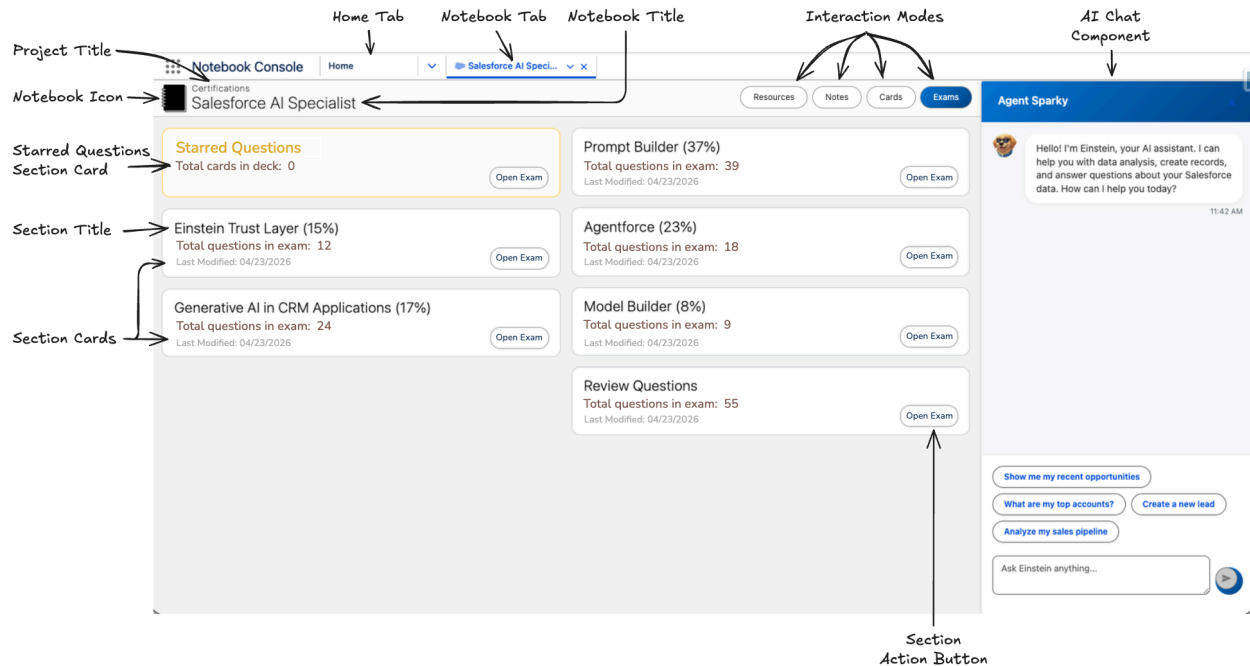
This is the third and final page of the 'Cards Interaction Mode - Upsert' page series. The purpose of this screen is to add related items from on the 'Resources Interaction Mode - List View' page for reference. Resources will be selected from the 'Resource Selector' component and added to the 'Resource List' component where items can be sorted using the 'Resource List Sort Buttons'. Clicking the 'Cancel' button will cancel adding/updating the card and navigate the user back to the 'Cards Interaction Mode - Details View' page. Clicking the back button will navigate the user to the previous page of this upsert series. Clicking the 'Save' button will save this card and navigate the user to the 'Cards Interaction Mode - Details View' page.

Cards Interaction Mode - Card Sort View



This is the 'Cards Interaction Mode - Card Sort View' page. The purpose of this screen is to allow the user to manually sort the card items in a specific order. Each card in the deck will be displayed in the 'Card List' and the user will be allowed to use the 'Card List Sort Buttons' to rearrange the order of the cards. Clicking the 'Cancel' button will cancel updating the card sort and navigate the user back to the 'Cards Interaction Mode - Details View' page. Clicking the 'Save' button will save the card sort and navigate the user to the 'Cards Interaction Mode - Details View' page.

Exam Interaction Mode - List View



Clicking the 'Exam' interaction mode button displays the 'Exam Interaction Mode - List View' page. The main portion consists of one or more 'Section Cards', each displaying the section's title along with details of the section such as total number of questions in the exam and last modified date. Each 'Section Card' (including the 'Starred Questions Section Card') will have an 'Open Exam' button in the bottom right corner of that section. Clicking the 'Open Exam' button will navigate the user to the 'Exam Interaction Mode - Details View' page. On the far right is the AI chat panel containing the Fetch Agent. The main purpose of this interaction mode is to provide an interface for organizing cards about a specific topic related to the notebook. There are no modals for this page.

Exam Interaction Mode - Details View

Example of Exam Question (Unsubmitted Answer):

This screenshot illustrates the 'Exam Interaction Mode - Details View' for an unsubmitted question. The interface includes a top navigation bar with tabs for 'Home', 'Notebook Tab', and 'Notebook Title'. Below this is a 'Project Title' section with a 'Notebook Console' and a 'Notebook Icon' for 'Salesforce AI Specialist'. A 'Notebook Section Selector' shows 'Einstein Trust Layer (15%)'. The main content area displays 'Question 6 of 39' with a 'Question Index' and 'Action Buttons (Add, Edit, Sort, Delete)'. The question text is: 'Global Insurance would like to ensure that the payment history of customers is available within Service Cloud for the support agents' reference. There are over 10 million records related to payment history currently stored in an external system. Cases that involve payment enquiries are less than 10% of the overall case volume. Which solution would a Salesforce Service Cloud Consultant recommend to provide the required access to support agents in Salesforce? Choose 1 answer.' The available answers are: A. Create a batch integration to import the payment history records into Salesforce every day, updating the related list on the customer detail page; B. Create a Visualforce page that retrieves payment history via a web service call; C. Due to the low volume of calls, building an integration is not cost-effective; D. Create a real-time integration so that the external system creates and updates payment history records in Salesforce. A 'Submit Button' is located at the bottom left. On the right, there is an 'AI Chat Component' with 'Agent Sparky' and a chat window. At the bottom, there are 'Exam Navigation Buttons' and a 'Close Button'.

Example of Exam Question (Submitted Answer):

This screenshot illustrates the 'Exam Interaction Mode - Details View' for a submitted question. The interface is similar to the previous one, but the question text is: 'Cosmic Software Solutions uses Salesforce for case management. When a case is closed in Salesforce, information about the issue in the case and the related solution should be sent securely to an external application that is used to maintain a repository of issues and solutions. The external application runs in an AWS Virtual Private Cloud (VPC). Exposure to external security threats should be minimized when sending the information to the external application. In addition to an HTTP callout that is automatically invoked from a flow when the status of a case changes, which of the following should be utilized to meet this requirement? Choose 1 answer.' The available answers are: A. Salesforce Connect; B. Shield Platform Encryption; C. Private Connect; D. External Services. The 'Selected Answer' is 'C. Private Connect', which is highlighted in green. Below the answers, there is an 'Answer with Explanation and Related Resources' section. The explanation states: 'Private Connect can be set up to route the HTTP(S) traffic between Salesforce and the external application through a fully managed connection that reduces the exposure to external security threats. It uses a feature called AWS PrivateLink to route the requests securely. Salesforce Connect is used to make external data available in Salesforce. It is not used to secure outbound requests. External Services is a feature that is'. The 'Close Button' is located at the bottom right. The 'AI Chat Component' and 'Exam Navigation Buttons' are also present.

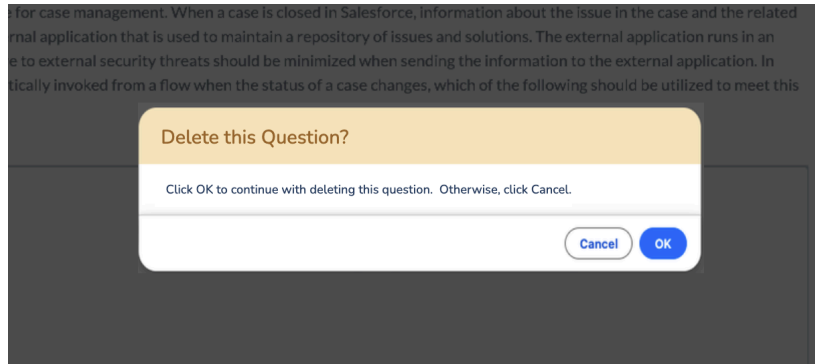
The 'Exam Interaction Mode - Details View' displays exam questions one at a time, used for learning information featuring a question and, once an answer is submitted, the corresponding answer with explanation and related resources will be displayed. The top of the page will display the notebook's section selector, allowing the user to jump to another section easily. To the right of the notebook section selector are four action buttons: Add (adding a new question), Edit (editing an existing question), Sort (defining the sort order of the questions) and Delete (for deleting a question). A 'Close' button on the lower right will allow the user to close this view and go back to the 'Cards Interaction Mode - List View' page. Finally, the bottom center will have two buttons for navigating through the exam questions. On the far right side is the AI chat panel containing the

Fetch Agent.

Clicking the 'Add' or 'Edit' action buttons will initiate the 'Exam Interaction Mode - Upsert View' series of pages.

Clicking the 'Sort' action button will initiate the 'Exam Interaction Mode - Question Sort View' page.

Clicking the 'Delete' action button will initiate the 'Delete Question Confirmation' modal shown below:



Exam Interaction Mode - Upsert Series (Page 1 of 2)

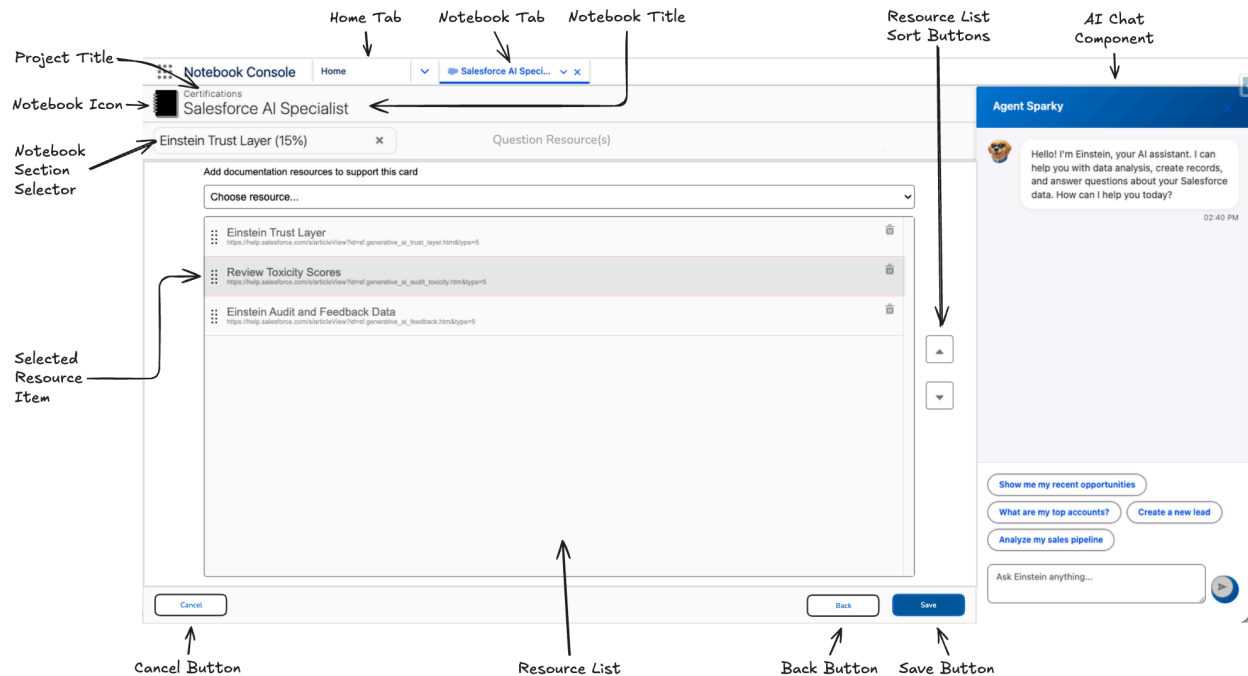
The screenshot displays the 'Exam Interaction Mode - Upsert Series' interface. The top navigation bar includes a 'Home Tab', 'Notebook Tab', 'Notebook Title', and 'Question Index'. The main content area shows a 'Project Title' (Salesforce AI Specialist), a 'Notebook Icon', and a 'Notebook Section Selector' (Einstein Trust Layer (15%)). The 'Exam Question' is displayed, followed by 'Available Answer Candidates' (Answer #1: Salesforce Connect, Answer #2: Shield Platform Encryption, Answer #3: Private Connect, Answer #4: External Services). The 'Actual Answer Selectors' are shown, with Answer #3 (Private Connect) selected. The 'Answer Summary' is provided below the candidates. The 'Cancel Button' is located at the bottom left, and the 'Next Button' is at the bottom right. An 'AI Chat Component' (Agent Sparky) is visible on the right side of the screen.

Annotations on the screenshot include:

- Project Title
- Notebook Icon
- Notebook Section Selector
- Exam Question
- Available Answer Candidates
- Actual Answer Selectors
- Answer Summary
- Cancel Button
- Next Button
- Home Tab
- Notebook Tab
- Notebook Title
- Question Index
- AI Chat Component

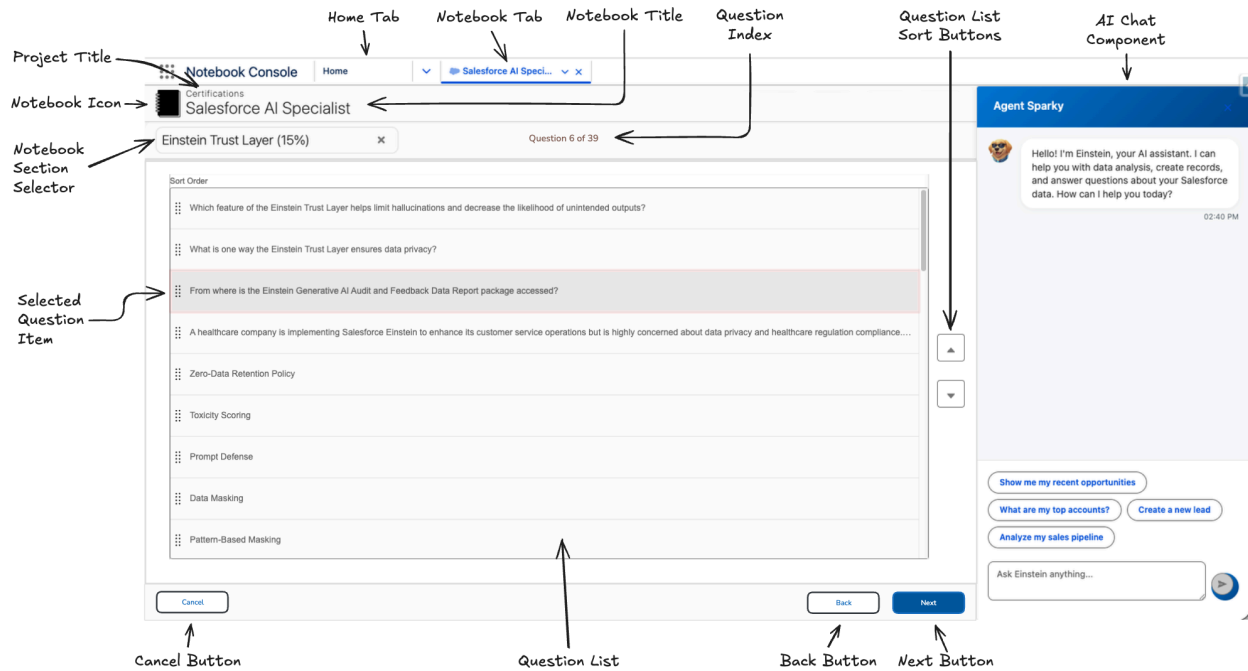
This is the first page of the 'Exam Interaction Mode - Upsert' page series. This screen captures the question, available answer candidates, actual answer selection and answer summary. Clicking the 'Cancel' button will cancel adding/updating the card and navigate the user back to the 'Cards Interaction Mode - Details View' page. Clicking the 'Next' button will navigate the user to the next page of the 'Cards Interaction Mode - Upsert Series (Page 2 of 3)' page series.

Exam Interaction Mode - Upsert Series (Page 2 of 2)



This is the second and final page of the 'Exam Interaction Mode - Upsert' page series. The purpose of this screen is to add related items from on the 'Resources Interaction Mode - List View' page for reference. Resources will be selected from the 'Resource Selector' component and added to the 'Resource List' component where items can be sorted using the 'Resource List Sort Buttons'. Clicking the 'Cancel' button will cancel adding/updating the question and navigate the user back to the 'Exam Interaction Mode - Details View' page. Clicking the back button will navigate the user to the previous page of this upsert series. Clicking the 'Save' button will save this question and navigate the user to the 'Exam Interaction Mode - Details View' page.

Exam Interaction Mode - Question Sort View



This is the 'Exam Interaction Mode - Question Sort View' page. The purpose of this page is to allow the user to manually sort the exam questions into a specific order. Each question in the deck will be displayed in the 'Question List' and the user will be allowed to use the 'Question List Sort Buttons' to rearrange the order of the exam questions. Clicking the 'Cancel' button will cancel updating the card sort and navigate the user back to the 'Exam Interaction Mode - Details View' page. Clicking the 'Save' button will save the sorting order of the exam questions and navigate the user to the 'Exam Interaction Mode - Details View' page.

Object Schema:

Below is a listing of the recommended objects schema along with their fields:

- **NSF Group:**
 - Object Name: NSF_Group
 - Description: The root object containing the collection of Groups.
 - Fields:
 - Name:
 - Field Name: Name
 - Type: Auto Number
 - Description: A unique auto-number field with the display format of G-{0000}
 - Title:
 - Field Name: Title
 - Type: Text(120)
 - Description: The title of the Group
- **NSF Project:**
 - Object Name: NSF_Project
 - Description: The collection of Projects which are children of Groups.
 - Fields:
 - Name:
 - Field Name: Name
 - Type: Auto Number
 - Description: A unique auto-number field with the display format of P-{0000}
 - Group:
 - Field Name: NSF_GroupId
 - Type: Lookup(NSF Group)
 - Description: A lookup field to the NSF_Group object
 - Title:
 - Field Name: Title
 - Type: Text(120)
 - Description: The title of the Project
- **NSF Notebook:**
 - Object Name: NSF_Notebook
 - Description: The collection of Notebooks which are children of Projects.
 - Fields:
 - Name:
 - Field Name: Name
 - Type: Auto Number
 - Description: A unique auto-number field with the display format of NB-{0000}
 - Project:
 - Field Name: NSF_ProjectId
 - Type: Lookup(NSF Project)
 - Description: A lookup field to the NSF_Project object
 - Title:
 - Field Name: Title
 - Type: Text(120)
 - Description: The title of the Notebook
- **NSF Section:**
 - Object Name: NSF_Section
 - Description: The collection of Sections which are children of Notebooks.
 - Fields:
 - Name:
 - Field Name: Name
 - Type: Auto Number

- Description: A unique auto-number field with the display format of S-{0000}
 - Notebook:
 - Field Name: NSF_NotebookId
 - Type: Lookup(NSF Notebook)
 - Description: A lookup field to the NSF_Notebook object
 - Title:
 - Field Name: Title
 - Type: Text(120)
 - Description: The title of the Section
- **NSF Resource:**
 - Object Name: NSF_Resource
 - Description: The collection of Resource feature items, which are children of Sections.
 - Fields:
 - Link:
 - Field Name: Link
 - Type: Long Text Area(1024)
 - Description: A link to a URL
 - Name:
 - Field Name: Name
 - Type: Auto Number
 - Description: A unique auto-number field with the display format of R-{0000}
 - Section:
 - Field Name: NSF_SectionId
 - Type: Lookup(NSF Section)
 - Description: A lookup field to the NSF_Section object
 - Title:
 - Field Name: Title
 - Type:
 - Description: The page title of the url defined in the Link field.
 - Type:
 - Field Name: Type
 - Type: Text(120)
 - Description: String describing the type of Resource. Options include 'Link', 'PDF' or 'Video'.
- **NSF Note:**
 - Object Name: NSF_Note
 - Description: The collection of Note feature items, which are children of Sections.
 - Fields:
 - Link:
 - Field Name: Link
 - Type: Long Text Area(1024)
 - Description: A link to a URL
 - Name:
 - Field Name: Name
 - Type: Auto Number
 - Description: A unique auto-number field with the display format of N-{0000}
 - Section:
 - Field Name: NSF_SectionId
 - Type: Lookup(NSF Section)
 - Description: A lookup field to the NSF_Section object
 - Title:
 - Field Name: Title

- Type: Text(120)
 - Description: The title of the Note
- **NSF Resource Note:**
 - Object Name: NSF_Resource_Note
 - Description: A junction object linking a N:N relationship between NSF_Resource and NSF_Note objects.
 - Fields:
 - Name:
 - Field Name: Name
 - Type: Auto Number
 - Description: A unique auto-number field with the display format of RN-{0000}
 - Note:
 - Field Name: NSF_NoteId
 - Type: Master-Detail(NSF Note)
 - Allow reparenting: True
 - Description: A lookup field to the NSF_Note object
 - Resource:
 - Field Name: NSF_ResourceId
 - Type: Master-Detail(NSF Resource)
 - Allow reparenting: True
 - Description: A lookup field to the NSF_Resource object
- **NSF Card:**
 - Object Name: NSF_Card
 - Description: The collection of Card feature items, which are children of Sections.
 - Fields:
 - Back:
 - Field Name: Back
 - Type: Long Text Area(131072)
 - Description: The backside content of the card
 - Front:
 - Field Name: Front
 - Type: Long Text Area(16448)
 - Description: The frontside content of the card
 - Name:
 - Field Name: Name
 - Type: Auto Number
 - Description: A unique auto-number field with the display format of C-{0000}
 - Section:
 - Field Name: NSF_SectionId
 - Type: Lookup(NSF Section)
 - Description: A lookup field to the NSF_Section object
 - Order:
 - Field Name: Order
 - Type: Number(8, 0)
 - Description: The numeric order of the card in relation to other cards in the same section.
- **NSF Resource Card:**
 - Object Name: NSF_Resource_Card
 - Description: A junction object linking a N:N relationship between NSF_Resource and NSF_Card objects.
 - Fields:
 - Card:
 - Field Name: NSF_CardId
 - Type: Master-Detail(NSF Card)
 - Allow reparenting: True
 - Description: A lookup field to the NSF_Note object

- Name:
 - Field Name: Name
 - Type: Auto Number
 - Description: A unique auto-number field with the display format of RC-{0000}
 - Resource:
 - Field Name: NSF_ResourceId
 - Type: Master-Detail(NSF Resource)
 - Allow reparenting: True
 - Description: A lookup field to the NSF_Resource object
- **NSF Exam:**
 - Object Name: NSF_Exam
 - Description: The collection of Exam feature items, which are children of Sections.
 - Fields:
 - Explanation:
 - Field Name: Explanation
 - Type: Long Text Area(1024)
 - Description: The explanation of the answer to the question
 - Name:
 - Field Name: Name
 - Type: Auto Number
 - Description: A unique auto-number field with the display format of E-{0000}
 - Section:
 - Field Name: NSF_SectionId
 - Type: Lookup(NSF Section)
 - Description: A lookup field to the NSF_Section object
 - Question:
 - Field Name: Question
 - Type: Long Text Area(1024)
 - Description: The exam question
- **NSF Exam Answer Candidates:**
 - Object Name: NSF_Exam_Answer_Candidates
 - Description: The collection of Exam answer candidates items, which are children of Exam.
 - Fields:
 - Possible Answer:
 - Field Name: Possible_Answer
 - Type: Long Text Area(1024)
 - Description: An answer candidate to a NSF_Exam record item.
 - Is Correct:
 - Field Name: Is_Correct
 - Type: Checkbox
 - Description: A true/false flag to determine if this record is an actual answer to an exam question.
 - Exam Item:
 - Field Name: NSF_Exam_ItemId
 - Type: Lookup(NSF Exam)
 - Description: A lookup field to the NSF_Exam object
- **NSF Resource Exam:**
 - Object Name: **NSF_Resource_Exam**
 - Description: A junction object linking a N:N relationship between NSF_Resource and NSF_Exam objects.
 - Fields:
 - Exam:
 - Field Name: NSF_ExamId
 - Type: Master-Detail(NSF Exam)
 - Allow reparenting: True

- Description: A lookup field to the NSF_Exam object
- Name:
 - Field Name: Name
 - Type: Auto Number
 - Description: A unique auto-number field with the display format of RE-{0000}
- Resource:
 - Field Name: NSF_ResourceId
 - Type: Master-Detail(NSF Resource)
 - Allow reparenting: True
 - Description: A lookup field to the NSF_Resource object
- **NSF Star Listing:**
 - Object Name: **NSF_Star_Listing**
 - Description: A reference object that contains lookups to feature objects.
 - Fields:
 - Name:
 - Field Name: Name
 - Type: Auto Number
 - Description: A unique auto-number field with the display format of SL-{0000}
 - Card:
 - Field Name: NSF_CardId
 - Type: Lookup(NSF Card)
 - Description: A lookup field to the NSF_Card object
 - Note:
 - Field Name: NSF_NoteId
 - Type: Lookup(NSF Note)
 - Description: A lookup field to the NSF_Note object
 - Resource:
 - Field Name: NSF_ResourceId
 - Type: Lookup(NSF Resource)
 - Description: A lookup field to the NSF_Resource object
 - Section:
 - Field Name: NSF_SectionId
 - Type: Lookup(NSF Section)
 - Description: A lookup field to the NSF_Section object

Technical Note (for Salesforce implementations only):

A custom modal component is included with the project to provide the application with the ability to prompt the user for details or notifications within a modal window. The name of this component is called 'lwcModal'. Attributes for this component include:

- class: The class name of the modal component so it can be referenced by the querySelector
- header: Text for the header of the modal
- confirm-button-label: Text for the confirmation button
- cancel-button-label: Text for the cancel button
- validate-on-confirm: Boolean to determine whether validation is on confirm
- error-message: An error message to display in case validation fails
- onbuttonclick: An event triggered function call

Sample implementation HTML code:

```
<c-lwc-modal
  class="upsertGroupModal"
  header={modalConfig.header}
  confirm-button-label={modalConfig.confirmButtonLabel}
  cancel-button-label={modalConfig.cancelButtonLabel}
  confirm-button-variant={modalConfig.confirmButtonVariant}
  validate-on-confirm=true
  error-message={modalConfig.errorMessage}
  focus-selector-string={modalConfig.focusSelectorString}
  onbuttonclick={handleUpsertGroupConfirm}
>
  <!-- Custom modal content -->
  <lightning-input
    tabindex="0"
    class="groupNameInput"
    data-id={modalConfig.id}
    data-name="groupName"
    value={modalConfig.value}
    label={modalConfig.label}
    required
  ></lightning-input>
</c-lwc-modal>
```